

1.0 Introduction

This report examines the process of extracting, analyzing, and scrubbing metadata from various file types using multiple tools, including ExifTool, FOCA, ExifCleaner, and PowerShell. Additionally, efforts were made to crack password-protected files using John the Ripper and Hashcat.

This revealed insights into encryption and security practices. The findings highlight the importance of metadata awareness, the effectiveness of different metadata removal tools, and the limitations of metadata scrubbing techniques.

1.1 Executive Summary

This report evaluates metadata extraction, analysis, and removal from various files. Tools such as ExifTool, FOCA, and ExifCleaner were used to extract metadata, while metadata removal was attempted with ExifTool, ExifPilot, and ExifCleaner.

Findings revealed that ExifTool effectively removed most metadata, but some fields (such as file system timestamps) remained immutable. Password-protected ZIP and PDF files were cracked using John the Ripper and Hashcat, with partial success.

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2.0 Exiftool metadata display with files

Command Used:

```
perl exiftool -r "C:\Users\N G\Desktop\metadata tools and data\Data" > metadata_full.txt
```

Findings:

- We can see a plethora of information about each file. FLAGS, such as User Access, Warning, MegaPixels, FocalLength, and plenty more are present in the text file.

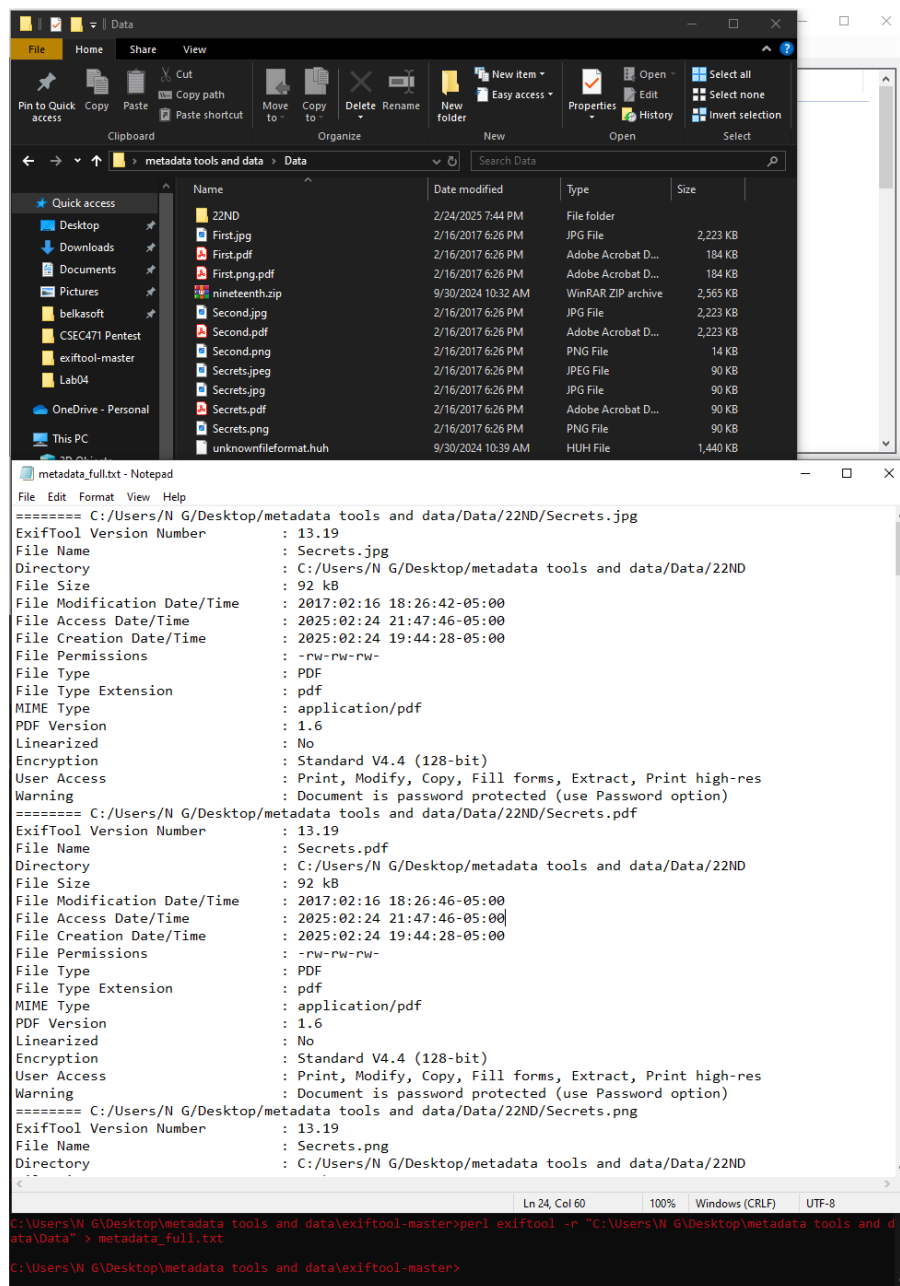


Image 2.0a: Displays metadata information gathered from exiftool.

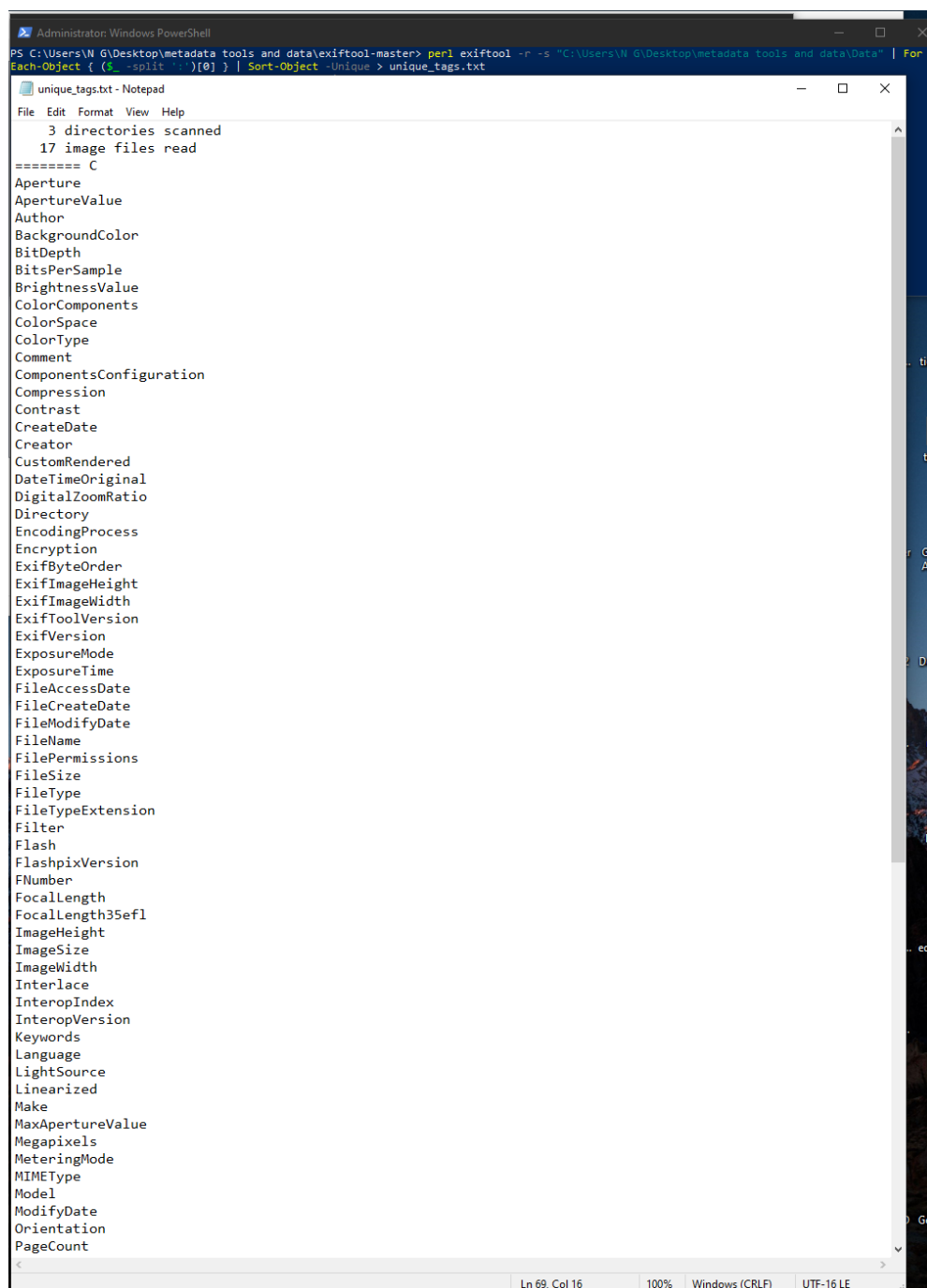
2.1 Exiftool unique TAGS extraction using powershell

Command Used (IN POWERSHELL):

```
powershell -command "perl exiftool -r -s "C:\Users\N G\Desktop\metadata tools and data\Data" | ForEach-Object { ($_ -split ':')[0] } | Sort-Object -Unique > unique_tags.txt"
```

Findings:

- By using this powershell command we have effectively cleaned up the metadata information in a way that only displays the TAGS without their value, essentially compiling the list of TAGS gathered without any duplicate rows. This will be used later to compare against how other tools handle metadata.



The screenshot shows two windows. The top window is an Administrator Windows PowerShell terminal. The command entered is: `perl exiftool -r -s "C:\Users\N G\Desktop\metadata tools and data\Data" | ForEach-Object { ($_ -split ':')[0] } | Sort-Object -Unique > unique_tags.txt`. The bottom window is a Notepad application titled 'unique_tags.txt - Notepad'. It displays the output of the command, which is a list of 117 unique metadata tags. The tags are listed in alphabetical order and include: Aperture, ApertureValue, Author, BackgroundColor, BitDepth, BitsPerSample, BrightnessValue, ColorComponents, ColorSpace, ColorType, Comment, ComponentsConfiguration, Compression, Contrast, CreateDate, Creator, CustomRendered, DateTimeOriginal, DigitalZoomRatio, Directory, EncodingProcess, Encryption, ExifByteOrder, ExifImageHeight, ExifImageWidth, ExifToolVersion, ExifVersion, ExposureMode, ExposureTime, FileAccessDate, FileCreateDate, FileModifyDate, FileName, FilePermissions, FileSize, FileType, FileTypeExtension, Filter, Flash, FlashpixVersion, FNumber, FocalLength, FocalLength35efl, ImageHeight, ImageSize, ImageWidth, Interlace, InteropIndex, InteropVersion, Keywords, Language, LightSource, Linearized, Make, MaxApertureValue, Megapixels, MeteringMode, MIMEType, Model, ModifyDate, Orientation, and PageCount.

```
File Edit Format View Help
3 directories scanned
17 image files read
===== C
Aperture
ApertureValue
Author
BackgroundColor
BitDepth
BitsPerSample
BrightnessValue
ColorComponents
ColorSpace
ColorType
Comment
ComponentsConfiguration
Compression
Contrast
CreateDate
Creator
CustomRendered
DateTimeOriginal
DigitalZoomRatio
Directory
EncodingProcess
Encryption
ExifByteOrder
ExifImageHeight
ExifImageWidth
ExifToolVersion
ExifVersion
ExposureMode
ExposureTime
FileAccessDate
FileCreateDate
FileModifyDate
FileName
FilePermissions
FileSize
FileType
FileTypeExtension
Filter
Flash
FlashpixVersion
FNumber
FocalLength
FocalLength35efl
ImageHeight
ImageSize
ImageWidth
Interlace
InteropIndex
InteropVersion
Keywords
Language
LightSource
Linearized
Make
MaxApertureValue
Megapixels
MeteringMode
MIMEType
Model
ModifyDate
Orientation
PageCount
```

Image 2.1a: Displays metadata TAG information gathered from the initial exiftool execution.

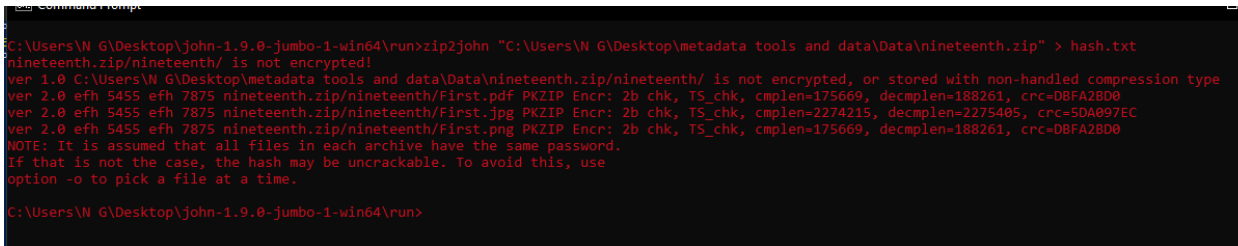
3.0 Johntheripper ZIP & PDF crack

Command Used:

zip2john "C:\Users\N G\Desktop\metadata tools and data\Data\nineteenth.zip" > hash.txt

Findings:

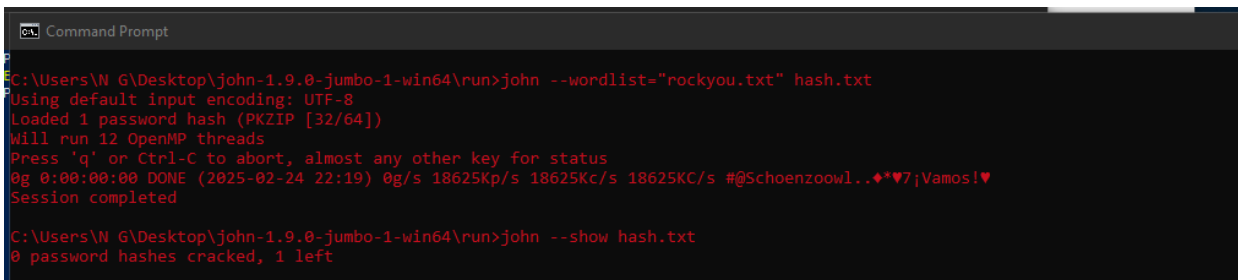
- After discovering that a zip file had password protection on it, I decided to try to crack it using johntheripper. The hash was obtained successfully. However, cracking the hash using the rockyou.txt wordlist did not prove successful. Likely due to the password not being included in rockyou.txt.



```
C:\Users\N G\Desktop\john-1.9.0-jumbo-1-win64\run>zip2john "C:\Users\N G\Desktop\metadata tools and data\Data\nineteenth.zip" > hash.txt
nineteenth.zip/nineteenth/ is not encrypted!
ver 1.0 C:\Users\N G\Desktop\metadata tools and data\Data\nineteenth.zip/nineteenth/ is not encrypted, or stored with non-handled compression type
ver 2.0 efh 5455 efh 7875 nineteenth.zip/nineteenth/First.pdf PKZIP Encr: 2b chk, TS_chk, cmplen=175669, decmplen=188261, crc=DBFA28D0
ver 2.0 efh 5455 efh 7875 nineteenth.zip/nineteenth/First.jpg PKZIP Encr: 2b chk, TS_chk, cmplen=2274215, decmplen=2275485, crc=50A097EC
ver 2.0 efh 5455 efh 7875 nineteenth.zip/nineteenth/First.png PKZIP Encr: 2b chk, TS_chk, cmplen=175669, decmplen=188261, crc=DBFA28D0
NOTE: It is assumed that all files in each archive have the same password.
If that is not the case, the hash may be uncrackable. To avoid this, use
option -o to pick a file at a time.

C:\Users\N G\Desktop\john-1.9.0-jumbo-1-win64\run>
```

Image 3.0a: Displays johntheripper obtaining hash of the ZIP file password.



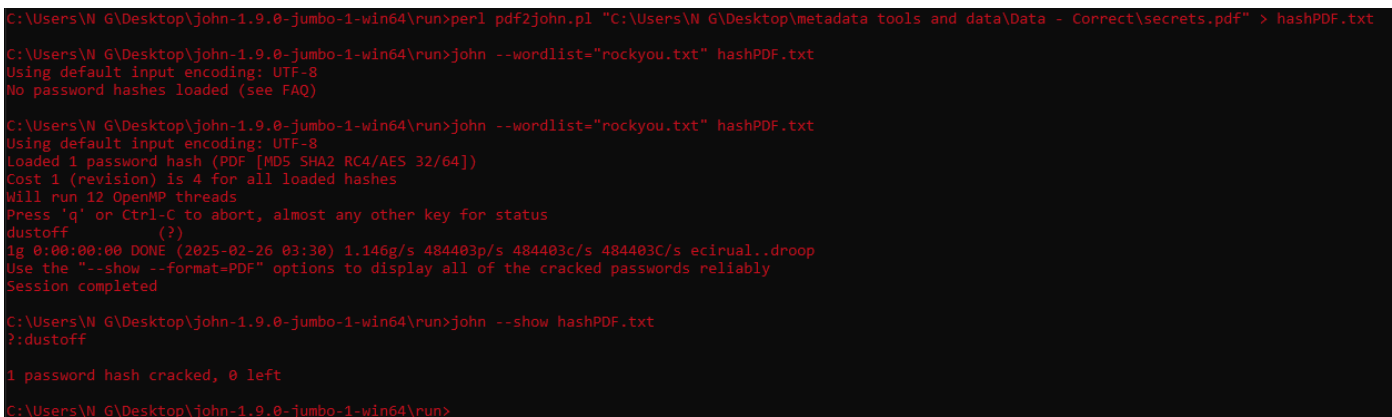
```
C:\Users\N G\Desktop\john-1.9.0-jumbo-1-win64\run>john --wordlist="rockyou.txt" hash.txt
Using default input encoding: UTF-8
Loaded 1 password hash (PKZIP [32/64])
Will run 12 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
0g 0:00:00:00 DONE (2025-02-24 22:19) 0g/s 18625Kp/s 18625Kc/s 18625Kc/s #@Schoenzoowl..♦♥7¡Vamos!♥
Session completed

C:\Users\N G\Desktop\john-1.9.0-jumbo-1-win64\run>john --show hash.txt
0 password hashes cracked, 1 left
```

Image 3.0b: Displays result of attempting to crack hash using rockyou.txt wordlist.

Findings:

- Alternatively, other PDFs were found to contain password locks on them too. There was success in finding the hash and cracked password “dustoff”. This allows us to open and view the content of the PDF file as shown below.



```
C:\Users\N G\Desktop\john-1.9.0-jumbo-1-win64\run>perl pdf2john.pl "C:\Users\N G\Desktop\metadata tools and data\Data - Correct\secrets.pdf" > hashPDF.txt
C:\Users\N G\Desktop\john-1.9.0-jumbo-1-win64\run>john --wordlist="rockyou.txt" hashPDF.txt
Using default input encoding: UTF-8
No password hashes loaded (see FAQ)

C:\Users\N G\Desktop\john-1.9.0-jumbo-1-win64\run>john --wordlist="rockyou.txt" hashPDF.txt
Using default input encoding: UTF-8
Loaded 1 password hash (PDF [MD5 SHA2 RC4/AES 32/64])
Cost 1 (revision) is 4 for all loaded hashes
Will run 12 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
dustoff
1g 0:00:00:00 DONE (2025-02-26 03:30) 1.146g/s 484403p/s 484403c/s 484403C/s ecirual..droop
Use the "--show --format=PDF" options to display all of the cracked passwords reliably
Session completed

C:\Users\N G\Desktop\john-1.9.0-jumbo-1-win64\run>john --show hashPDF.txt
?:dustoff

1 password hash cracked, 0 left

C:\Users\N G\Desktop\john-1.9.0-jumbo-1-win64\run>
```

Image 3.0c: Displays johntheripper obtaining hash of the PDF file password, and successful crack.

Findings:

- The contents of the file seem to indicate that the files themselves are identical. This is later confirmed in section 6.0. Additionally, these files extension names were changed to PDF as that was their true file type. “Secrets.pdf” was “Secrets.jpg”, “Secrets1.pdf” was “Secrets.pdf”, and “Secrets2.jpg” was “Secrets.png”.

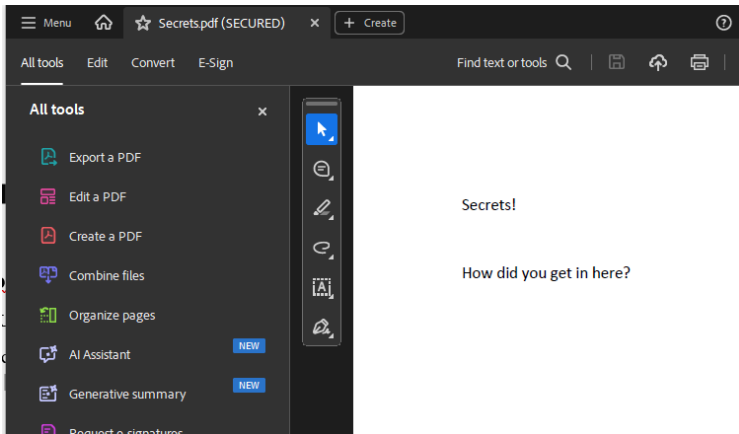


Image 3.0d: Displays content of locked PDF file “Secrets.pdf” which was “Secrets.jpg”.

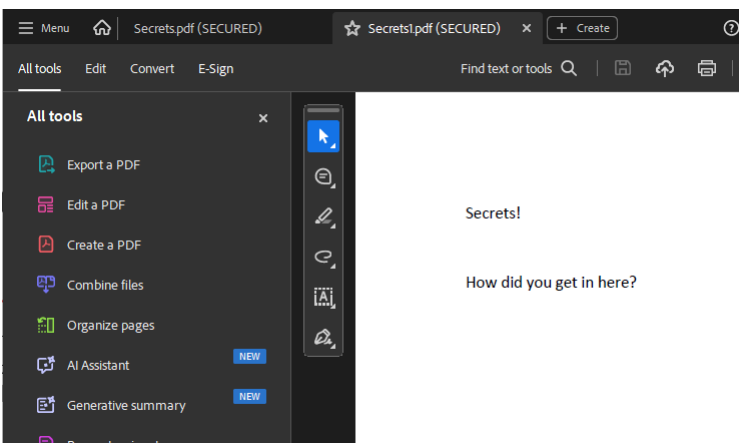


Image 3.0e: Displays content of locked PDF file “Secrets1.pdf” which was “Secrets.pdf”.

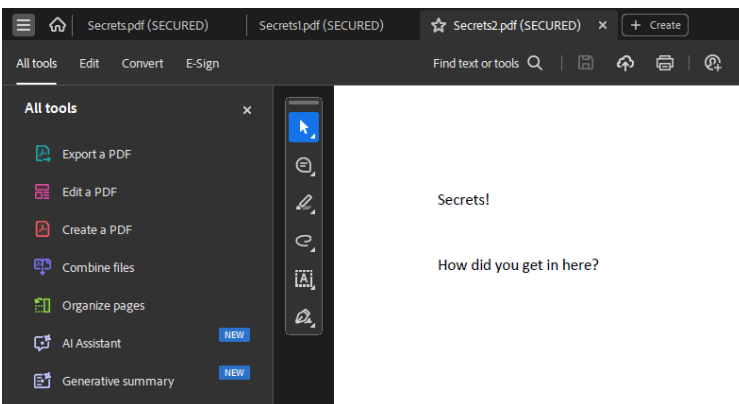


Image 3.0f: Displays content of locked PDF file “Secrets2.pdf” which was “Secrets.png”.

4.0 FOCA

Using FOCA we can see metadata information displayed out in a GUI type fashion. We can see TAGS such as Flash, Focal Length, Shutter Speed Value, and many others. This is a nicer way to view multiple files as the data is organized into folders similar to how file explorer shows data.

Findings:

- Using FOCA we can see similar results like exiftool however in a GUI format. We can see tags such as Aperture Value, Flash, and many others. Additionally, we have provided to us interesting information in the MetaData Summary Tab with Users, and Software being populated with information.

The screenshot displays the FOCA (final version) 3.4 application window. The interface is divided into several sections:

- Project Tree (Left):** A hierarchical view of the project structure, including folders like 'Documents (18/18)', 'Unknown (4)', and 'Metadata Summary'.
- Attribute-Value Table (Center):** A table listing various metadata attributes and their corresponding values for the selected file (Exif Makernote).
- Log (Bottom):** A table showing the results of the metadata extraction process, including time, source, severity, and message.

Attribute-Value Table Data:

Attribute	Value
Exif Makernote	
Make	Motorola
Model	DROID RAZR HD
Orientation	Top, left side (Horizontal / normal)
X Resolution	72 dots per inches
Y Resolution	72 dots per inches
Resolution Unit	Inches
Date/Time	2013:04:27 11:38:50
YCbCr Positioning	Center of pixel array
Exposure Time	34/50000 sec
F-Number	F 2.4
Exif Version	2.20
Date/Time Original	2013:04:27 11:38:50
Date/Time Digitized	2013:04:27 11:38:50
Components Configuration	YCbCr
Shutter Speed Value	1/-2147483648 sec
Aperture Value	F 2.4
Brightness Value	-2147483648
Max Aperture Value	F 2.4
Metering Mode	Average
Light source	23
Flash	Flash did not fire
Focal Length	4.35 mm
FlashPix Version	1.00
Color Space	sRGB
Exif Image Width	3264 pixels
Exif Image Height	2448 pixels
Scene Type	Directly photographed image
Custom Rendered	1
Exposure Mode	Auto bracket
White balance mode	Auto white balance
Digital Zoom Ratio	1
Scene Capture Type	Standard
Contrast	None
Saturation	None
Sharpness	Low
Compression	JPEG (old-style)
Thumbnail Offset	958 bytes
Thumbnail Length	26375 bytes
Thumbnail Data	[26375 bytes of thumbnail data]

Exif Interoperability Makernote:

Attribute	Value
Interoperability Index	Recommended Exif Interoperability Rules (ExifR98)
Interoperability Version	1.00

Log Table:

Time	Source	Severity	Message
11:59:53...	MetadataSearch	low	Document metadata extracted: C:\Users\N G\Desktop\metadata tools and data\Data - FOCA\Secre...
11:59:53...	MetadataSearch	low	Document metadata extracted: C:\Users\N G\Desktop\metadata tools and data\Data - FOCA\Seco...
11:59:53...	MetadataSearch	low	Document metadata extracted: C:\Users\N G\Desktop\metadata tools and data\Data - FOCA\22ND...
11:59:53...	MetadataSearch	low	Document metadata extracted: C:\Users\N G\Desktop\metadata tools and data\Data - FOCA\Secre...
11:59:53...	MetadataSearch	low	Document metadata extracted: C:\Users\N G\Desktop\metadata tools and data\Data - FOCA\Secre...
11:59:53...	MetadataSearch	low	Document metadata extracted: C:\Users\N G\Desktop\metadata tools and data\Data - FOCA\22ND...
11:59:53...	MetadataSearch	low	Document metadata extracted: C:\Users\N G\Desktop\metadata tools and data\Data - FOCA\22ND...
11:59:53...	MetadataSearch	low	Document metadata extracted: C:\Users\N G\Desktop\metadata tools and data\Data - FOCA\Secre...
11:59:53...	MetadataSearch	low	Document metadata extracted: C:\Users\N G\Desktop\metadata tools and data\Data - FOCA\Seco...
11:59:53...	MetadataSearch	low	Document metadata extracted: C:\Users\N G\Desktop\metadata tools and data\Data - FOCA\22ND...
11:59:53...	MetadataSearch	low	Document metadata extracted: C:\Users\N G\Desktop\metadata tools and data\Data - FOCA\22ND...
11:59:53...	MetadataSearch	low	Document metadata extracted: C:\Users\N G\Desktop\metadata tools and data\Data - FOCA\22ND...
11:59:53...	MetadataSearch	low	Document metadata extracted: C:\Users\N G\Desktop\metadata tools and data\Data - FOCA\First.jpg
11:59:53...	MetadataSearch	low	Document metadata extracted: C:\Users\N G\Desktop\metadata tools and data\Data - FOCA\First.p...
11:59:53...	MetadataSearch	low	Document metadata extracted: C:\Users\N G\Desktop\metadata tools and data\Data - FOCA\First.pdf

Image 4.0a: Displays metadata information via FOCA's GUI.

5.0 HxD, and POWERSHELL

During the exercise it was observed that there was a file with a .huh as an extension. Numerous tries to rename the file to .jpg, .jpeg, .pdf, .docx, etc... failed to yield any positive results. Powershell and HxD were used to find the “first 10 bytes (magic bytes)” that may contain signatures as to what file format they belong to.

Findings:

- Using Powershell and HxD we observed the first bytes in the file. Trying to google or look up on any of the signature websites (such as www.garykessler.net/library/file_sigs.html) did not yield any helpful results.

Command Used (IN POWERSHELL):

Get-Content "C:\Users\N G\Desktop\metadata tools and data\Data\unknownfileformat.huh" -First 10 | Format

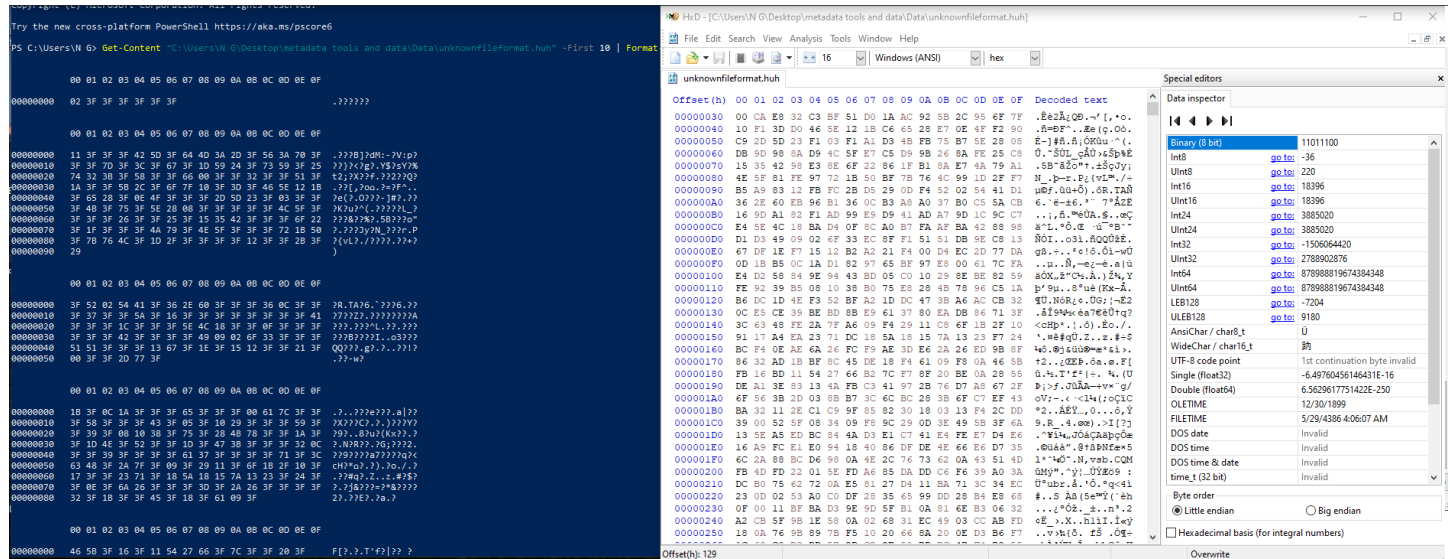


Image 5.0a: *Displays hexcode data of unknownfileformat.huh using Powershell and HxD.*

6.0 Metadata removal findings

Upon erasing the metadata information, I realized that some files had wrong file extension names. After changing the file extension names to PDF it was discovered they were password protected.

Findings:

- Using Powershell and HxD we observed the first bytes in the file. Trying to google, or look up on any of the signature websites (such as www.garykessler.net/library/file_sigs.html) did not yield any helpful results.

Command Used (IN POWERSHELL):

Get-Content "C:\Users\N G\Desktop\metadata tools and data\Data\unkownfileformat.huh" -First 10 | Format

```
Command Prompt
Microsoft Windows [Version 10.0.19045.5487]
(c) Microsoft Corporation. All rights reserved.

C:\Users\N G>exiftool
'exiftool' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\N G>cd "C:\Users\N G\Desktop\metadata tools and data\exiftool-master"

C:\Users\N G\Desktop\metadata tools and data\exiftool-master>perl exiftool -All= -overwrite_original "C:\Users\N G\Desktop\metadata tools and data\ALLFILES"
Error: File not found - C:\Users\N G\Desktop\metadata tools and data\ALLFILES
  0 image files updated
  1 files weren't updated due to errors

C:\Users\N G\Desktop\metadata tools and data\exiftool-master>perl exiftool -All= -overwrite_original "C:\Users\N G\Desktop\metadata tools and data\Data - Copy"
Warning: [minor] Exiftool PDF edits are reversible. Deleted tags may be recovered! - C:\Users\N G\Desktop\metadata tools and data\Data - Copy\First.pdf
Warning: [minor] Exiftool PDF edits are reversible. Deleted tags may be recovered! - C:\Users\N G\Desktop\metadata tools and data\Data - Copy\First.png.pdf
Error: Not a valid PDF (looks more like a JPEG) - C:\Users\N G\Desktop\metadata tools and data\Data - Copy\Second.pdf
Error: Not a valid JPEG (looks more like a PDF) - C:\Users\N G\Desktop\metadata tools and data\Data - Copy\Secrets.jpeg
Error: Not a valid JPG (looks more like a PDF) - C:\Users\N G\Desktop\metadata tools and data\Data - Copy\Secrets.jpg
Error: Document is password protected (use Password option) - C:\Users\N G\Desktop\metadata tools and data\Data - Copy\Secrets.pdf
Error: Not a valid PNG (looks more like a PDF) - C:\Users\N G\Desktop\metadata tools and data\Data - Copy\Secrets.png
  1 directories scanned
  5 image files updated
  5 files weren't updated due to errors

C:\Users\N G\Desktop\metadata tools and data\exiftool-master>
```

Image 6.0a: Displays erasing metadata and associated warnings.

Findings:

Using MD5 hash on the renamed files we verify the uniqueness of the files. It turns out that they are the exact same file. It might be safe to assume that they all share the same password.

The screenshot shows a web browser window with the 'MD5 File Hash Online Calculator' website. On the left, a Windows File Explorer window is open, showing a folder named 'metadata tools and data' containing files like '22ND', 'First.jpg', 'First.pdf', 'First.png.pdf', 'nineteenth.zip', 'Second.jpeg', 'Second.png', 'Secrets.pdf', 'Secrets.png', 'Secrets1.pdf', 'Secrets2.pdf', and 'unknownfileformat'. The calculator interface has a 'Drop files here or click to select' area. Below it, three file entries are listed, each with a size of 92041 bytes:

- Secrets2.pdf** - 92041 bytes
MD5: 109cb619faf566c7ae4dd027d3cae3fb
SHA1: f4ef2c926c781cf91b23ad8dd64eeb70a2588ca
SHA256: aa2592e66b8cb1358a7d3c5ec759462581839f1d9a6d8ec375b88f240d12c5d3
SHA512: fea355979ffada5d4014273c78eb11b4e101f923902bafa164e68b402981309dd24258764e2d99fe7c1aa9c659e13e06b6cf2dd98996842124206a4d0363ed27
- Secrets1.pdf** - 92041 bytes
MD5: 109cb619faf566c7ae4dd027d3cae3fb
SHA1: f4ef2c926c781cf91b23ad8dd64eeb70a2588ca
SHA256: aa2592e66b8cb1358a7d3c5ec759462581839f1d9a6d8ec375b88f240d12c5d3
SHA512: fea355979ffada5d4014273c78eb11b4e101f923902bafa164e68b402981309dd24258764e2d99fe7c1aa9c659e13e06b6cf2dd98996842124206a4d0363ed27
- Secrets.pdf** - 92041 bytes
MD5: 109cb619faf566c7ae4dd027d3cae3fb
SHA1: f4ef2c926c781cf91b23ad8dd64eeb70a2588ca
SHA256: aa2592e66b8cb1358a7d3c5ec759462581839f1d9a6d8ec375b88f240d12c5d3
SHA512: fea355979ffada5d4014273c78eb11b4e101f923902bafa164e68b402981309dd24258764e2d99fe7c1aa9c659e13e06b6cf2dd98996842124206a4d0363ed27

Image 6.0b: Displays MD5 hash of the previously changed file extensions, confirming they are identical.

7.0 Metadata removal comparison exiftool

Here we will remove the metadata using exiftool and compare the TAGS to the previously found tags to see which TAGS have been removed.

Findings:

- The left text window contains the initial TAGS found by exiftool. The middle window contains the TAGS after removal. And our findings on the right window shows the TAGS which were removed.

Command Used (IN POWERSHELL):

```
perl exiftool -r -s "C:\Users\N_G\Desktop\metadata tools and data\Data" | ForEach-Object { ($_.split(':')[0]) | Sort-Object -Unique > tags_before.txt
```

```
perl exiftool -r -s "C:\Users\N_G\Desktop\metadata tools and data\Data Copy" | ForEach-Object { ($_.split(':')[0]) | Sort-Object -Unique > tags_after.txt
```

```
Compare-Object (Get-Content tags_before.txt) (Get-Content tags_after.txt) | Where-Object { $_.SideIndicator -eq <= } | Out-File removed_tags.txt
```

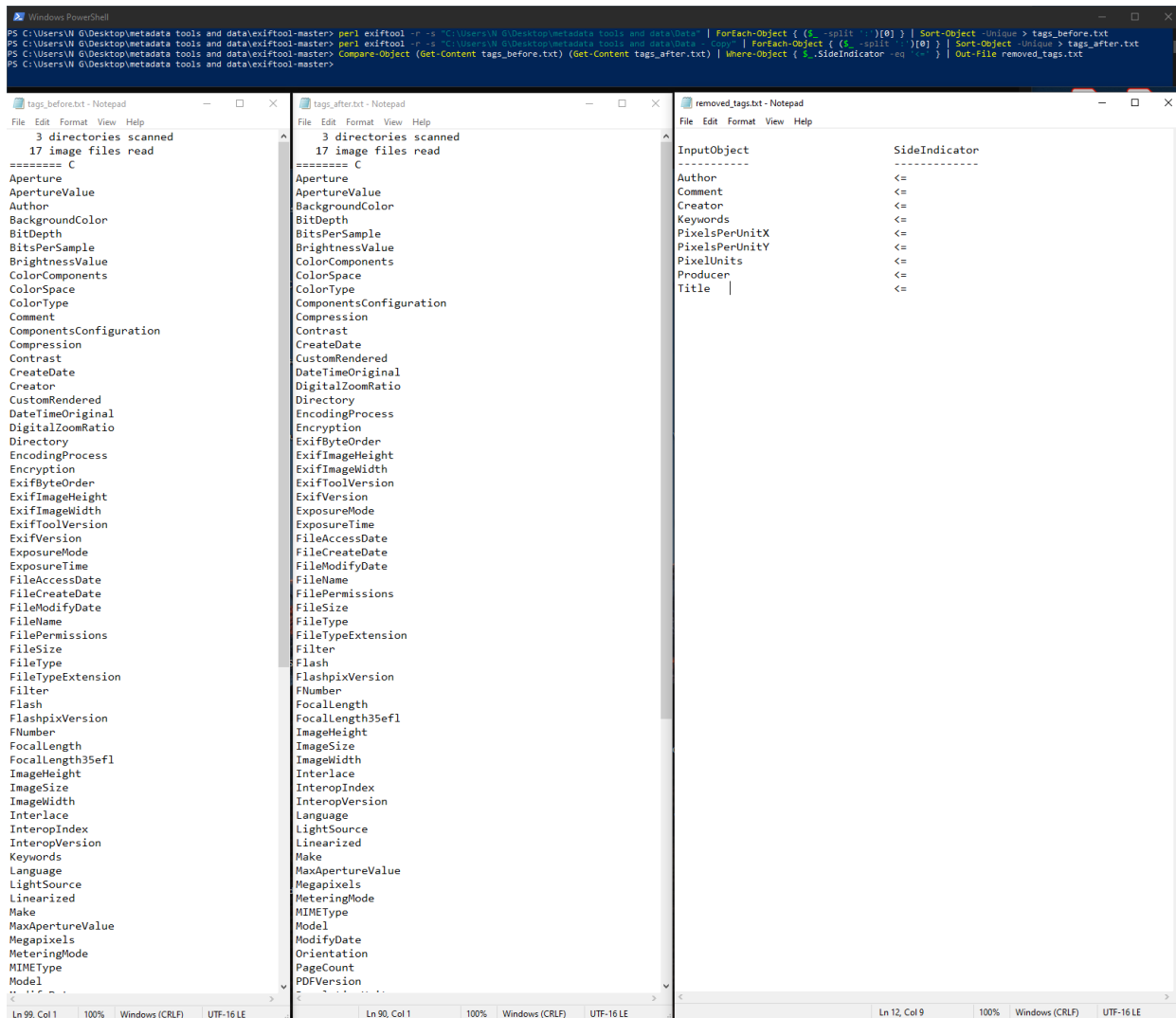


Image 7.0a: Displays before, after, and difference of Metadata removal using exiftool.

8.0 Metadata removal comparison ExifPilot

Here we will remove the metadata using ExifPilot and compare the TAGS to the previously found tags to see which TAGS have been removed.

Findings:

- The left text window contains the initial TAGS found by ExifPilot. The middle window contains the TAGS after removal. And our findings on the right window shows us the TAGS which were removed.

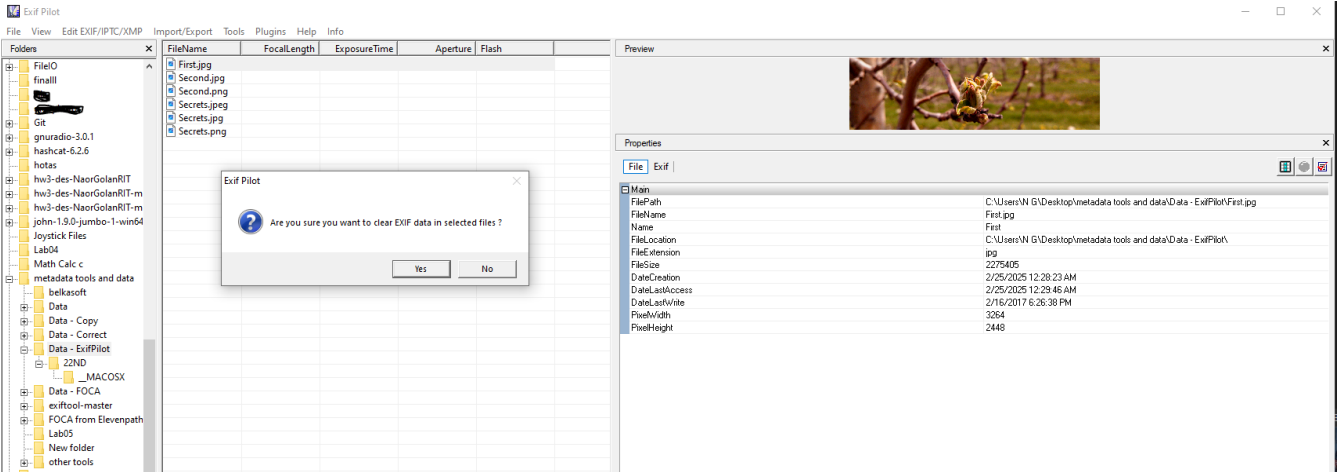


Image 8.0b: Displays before, after, and difference of Metadata removal using ExifPilot.

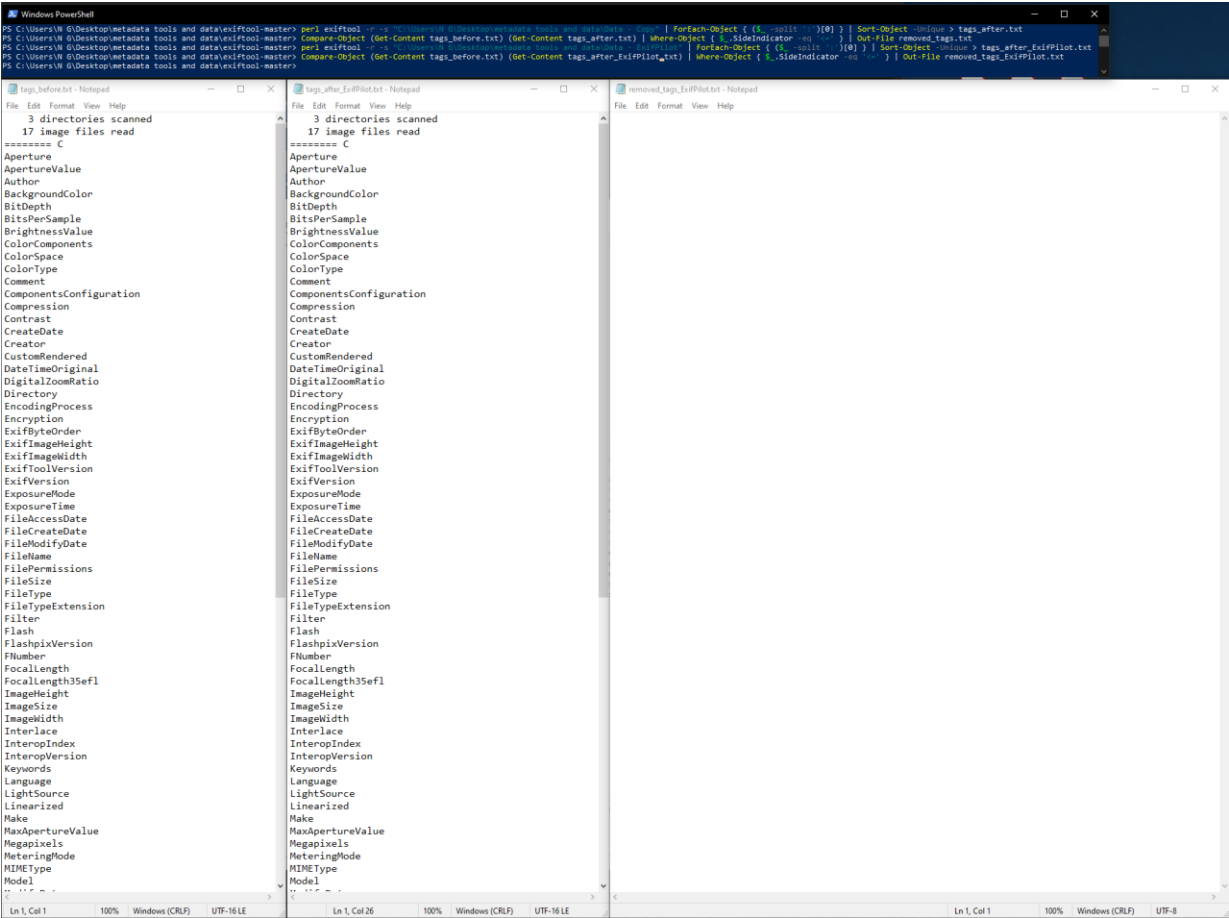


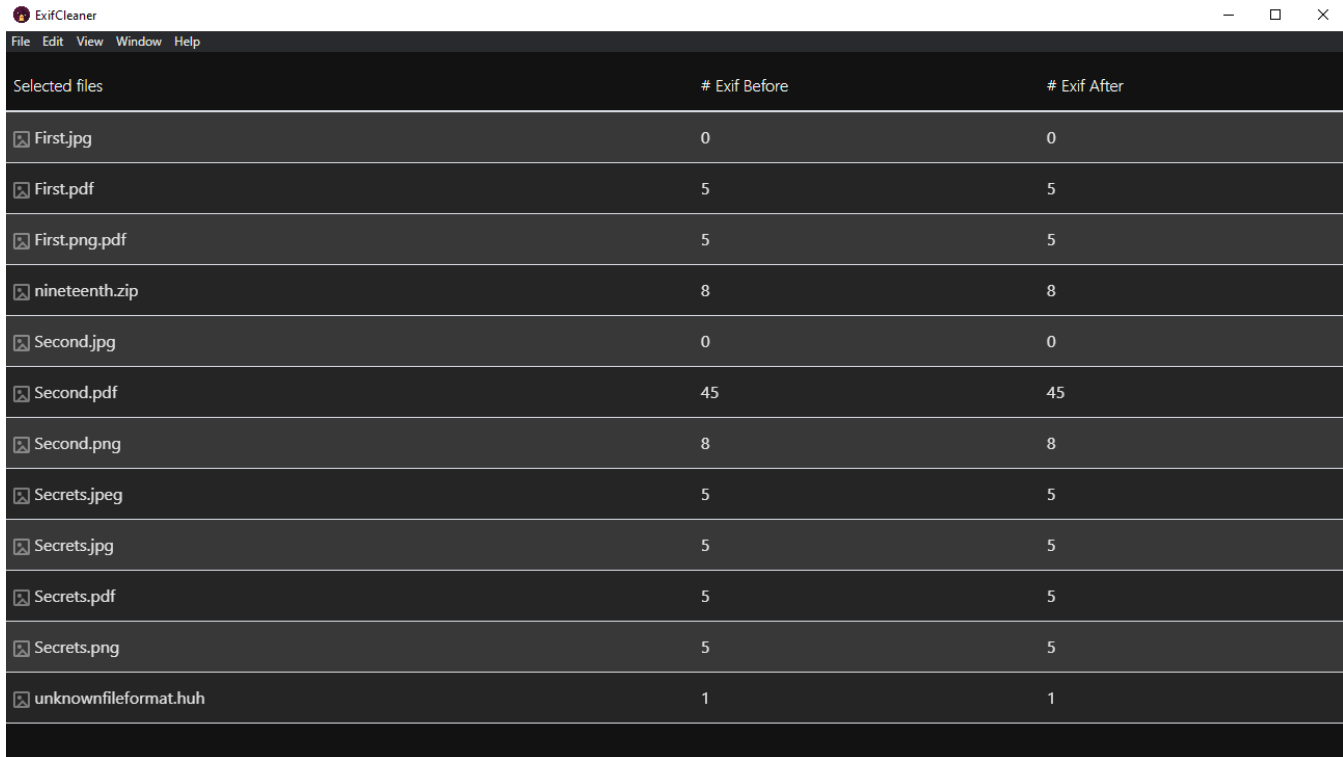
Image 8.0b: Displays before, after, and difference of Metadata removal using ExifPilot (was not able to remove data).

9.0 Metadata removal comparison ExifCleaner

Here we will remove the metadata using ExifCleaner and compare the TAGS to the previously found tags to see which TAGS have been removed.

Findings:

- The left text window contains the initial TAGS found by ExifCleaner. The middle window contains the TAGS after removal. And our findings on the right window shows us the TAGS which were removed.



The screenshot shows the ExifCleaner application window with a menu bar (File, Edit, View, Window, Help) and a table of files. The table has three columns: 'Selected files', '# Exif Before', and '# Exif After'. The files listed are First.jpg, First.pdf, First.png.pdf, nineteenth.zip, Second.jpg, Second.pdf, Second.png, Secrets.jpeg, Secrets.jpg, Secrets.pdf, Secrets.png, and unknownfileformat.huh. The counts for each file are as follows:

Selected files	# Exif Before	# Exif After
First.jpg	0	0
First.pdf	5	5
First.png.pdf	5	5
nineteenth.zip	8	8
Second.jpg	0	0
Second.pdf	45	45
Second.png	8	8
Secrets.jpeg	5	5
Secrets.jpg	5	5
Secrets.pdf	5	5
Secrets.png	5	5
unknownfileformat.huh	1	1

Image 9.0a: Displays the files being stripped of their metadata (there are more but only showing main folder).

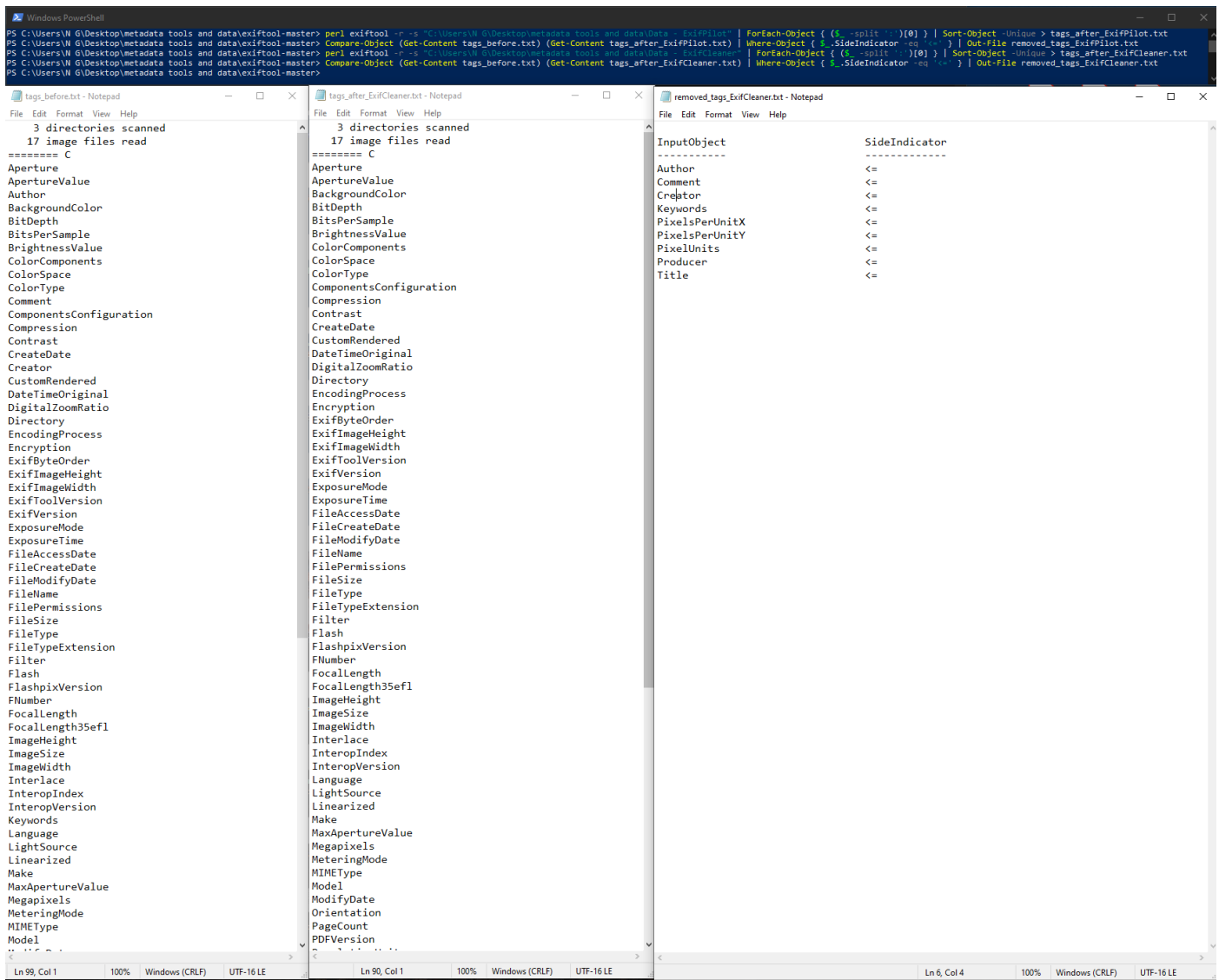


Image 9.0b: Displays before, after, and difference of Metadata removal using ExifCleaner.

10.0 Summary of metadata findings

Here is a table summary of what TAGS were mutable using our tools and methods.

Metadata TAG	Exiftool (removed)	ExifPilot (removed)	ExifCleaner (removed)
Author	Yes	No	Yes
Comment	Yes	No	Yes
Creator	Yes	No	Yes
Keywords	Yes	No	Yes
PixelsPerUnitX	Yes	No	Yes
PixelsPerUnitY	Yes	No	Yes
PixelUnits	Yes	No	Yes
Producer	Yes	No	Yes
Title	Yes	No	Yes
(the rest of the tags are not included as they were immutable).	NO	NO	NO

Business impact:

Metadata, when improperly handled, can lead to security vulnerabilities. The results of this lab show that metadata extraction tools can reveal sensitive information such as author names, creation timestamps, and GPS coordinates.

If not removed properly, metadata can expose:

- Corporate security risks (leaking internal document authorship).
- Privacy concerns (exposing location data from images).
- Forensic investigation trails (attackers removing metadata to erase digital footprints).

The importance of metadata management is crucial for organizations dealing with sensitive files. Regular metadata scrubbing should be implemented as part of data loss prevention (DLP) policies to reduce exposure risk.